

U.K.V.S

Universal Knight Vision Scope

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Chapter 2: Project Description

2.1 Motivation and Background

The optics industry offers a wide range of magnified and infrared imaging systems; however, many infrared-based devices are expensive and designed primarily for specific lighting conditions. This creates a limitation for users who may need an optic capable of functioning across a range of environments, from bright daylight to low-light and near-dark conditions. Potential applications for a versatile optical system include wildlife observation and photography, navigation through remote areas during evening hours, and defense-related imaging. A system capable of adapting to changing illumination conditions could reduce the need for separate optical devices for different environments.

The history of magnified optics dates back to 1608, when Dutch spectacle maker Hans Lippershey applied for a patent for an early telescope, commonly referred to as the Dutch telescope. Its basic optical design consisted of a convex objective lens and a concave eyepiece and provided approximately 3× magnification. In 1609, Galileo Galilei developed an improved version of the telescope, achieving substantially greater magnification and enabling more detailed astronomical observations. Since the development of the telescope, magnified optics have been adapted for numerous applications, including navigation, scientific observation, surveying, and hunting.

More than three centuries later, in 1943, Germany developed an early practical night-vision system known as the Zielgerät 1229, or “Vampir.” The system was cumbersome by modern standards, weighing approximately 5 lb, while its required power supply was carried by the user and weighed approximately 30 lb. Additionally, the system required an active infrared illumination source because it could not sufficiently detect and amplify naturally occurring ambient light. These limitations illustrate the challenges associated with early night-vision technology and the continuing need for optical systems that are compact, adaptable, and capable of operating across a broad range of lighting conditions.

Our motivation is to design a dual-purpose optic that enhances the user's ability to see further distances in the daytime as well as low light conditions so the user does not need to switch optic devices. This collaborative effort combines photonics, electrical engineering, and computer engineering principles to create a project that fully encompasses all fields involved in the project by incorporating a magnifying lens system, an IR intake system, an image processing and display system, and an efficient PCB design. This magnifying optic project has many tasks and opportunities for all four team members to further their

knowledge in their field while also enhancing their current skills and refining other needed skills that they may need in the future.

2.2 Existing products

Existing products in this field are typically not dual-purpose optics or cameras. These devices generally do either visible ambient light only or IR spectrum only.

- PVS-14 night vision monocular: This device is strictly a near IR optic. It has its own built in IR light source for especially low light conditions. It is powered by a single AA battery and has a switch with 3 options, off, on, and on with IR light. At the front you can manually focus the optic to which distance you wish to observe, generally between 5 to 50 yards. It has no magnification and is restricted to low light use. The device is under 8" long and has a fitting to mount to a helmet or head harness. The inexpensive version of this device uses a green phosphorous output so the image is varying shades of green. It is not useable in well lit conditions as visible light can potentially burn out the sensors.

This product relates to ours as it has a lens system that focuses onto a sensor which translates the image for the user. It also uses an external battery source which our system will require.

- 3x magnifiers: these devices offer a user a fixed magnification device. It requires zero power source and is easily focused between distances, usually between 5 to 200 yards. It is generally a very small device being less than 4-5" long. It has no IR abilities and is not usable in dark conditions without external attachments.

This device relates to ours as we are creating a device that has similar magnification.

- Edge 3 Cellular Game Camera. These cameras have a motion sensor and photo sensor built in. The motion sensor detects movement to tell the camera to take a still picture of the object (or animal) in front of it. The photo sensor determines ambient light settings which tells the camera to take a visible light image or an IR image of the object in front of it. The device also has wifi and ranged internet capabilities to send images to the user remotely. Drawbacks of the device for our project are that it does not have any telescoping or focusing capabilities; it is on a fixed plane with few options. It is also not a telescope in terms of a device that a

user can hold or prop up to look at an image at a distance in real time as the automation of the device leaves little user input on how the device operates or when it takes pictures.

This device relates to ours as it does have visible and IR capabilities inside the same unit. It uses a photo detection system to determine ambient light to determine if the image captured should be visible or IR.

- Trijicon VCOG: the VCOG is a fixed power magnified optic

2.3 Goals and Objectives

2.3.1 Basic goals

We intend to develop an optical device capable of operating across a range of ambient lighting conditions, eliminating the need for users to switch between multiple optical tools. The optic will utilize a single viewport that provides a magnified image, allowing users to observe distant objects with sufficient clarity to make informed observations. Under normal lighting conditions, the device will function as a conventional short-range telescope. As ambient light levels decrease, the system will transition to infrared imaging and project the resulting image through the eyepiece, allowing the user to maintain a consistent viewing interface without changing optical devices. The optic will autonomously monitor ambient lighting conditions and adjust display brightness accordingly. An integrated IR illuminator will provide additional infrared illumination in low-light environments to improve image quality.

The electrical system will be designed to optimize power consumption and support at least 12 hours of continuous operation. The device will include a master on/off switch with appropriate debouncing to prevent unintended switching and protect electrical components. A control algorithm will continuously monitor ambient light levels and generate a proportional control signal to automatically adjust the display brightness, increasing or decreasing the output based on the detected lighting conditions.

2.3.2 Advance goals

A lens focusing system will be implemented to allow the user to clearly view objects at varying distances. Although the optic will provide only 3–5× magnification, the ability to

adjust focus between different viewing distances will improve usability and image clarity. A forward-facing photodetector will measure ambient light levels and communicate this information to the command module, which will control the display brightness and determine when the IR illuminator should be activated or deactivated. The optical lens system will direct the image into a camera capable of detecting both visible and infrared wavelengths. One example of a suitable camera is the Raspberry Pi Global Shutter Camera, which utilizes a 1.6 MP Sony IMX296 sensor with sensitivity to both visible and near-infrared light. The camera includes an IR-cut filter that can be removed to enable infrared imaging for this design.

Electrical components will be cleanly integrated into the device housing, with appropriately sized PCBs designed to support the system's functionality while minimizing the overall footprint. The calibration procedure will be designed to be repeatable and minimize the need for manual adjustments to the alignment between the visible and IR optical systems. The device will also prioritize power efficiency and system stability, with control algorithms designed to prevent noticeable flickering during transitions between lighting conditions and display brightness levels.

2.3.4 Stretch goals

Potential stretch goal for the project includes integrating a rangefinder into the optic to provide the user with additional environmental information. The rangefinder would measure the distance to the observed object and display the measurement through the ocular interface.

Another potential stretch goal is adjustable magnification, allowing the user to manually vary the optical magnification based on the viewing distance. For example, an object at 100 meters may occupy too much of the field of view at 10× magnification, while an object at 1000 meters may be difficult to observe at only 3× magnification. Variable magnification would provide greater flexibility across a wider range of viewing distances.

Improving the infrared imaging system to achieve a response time of approximately one second would enable faster transitions between visible and IR imaging, providing a more seamless user experience. A manual override could also be incorporated to allow the user to switch between day and night modes independently of the automatic lighting system.

Rather than using a binary on/off transition between visible and IR imaging, the system could gradually adjust the contribution of the IR feed based on changes in ambient light

levels. This approach could provide a smoother transition between viewing modes and reduce abrupt changes in image appearance.

2.3.5 Objectives

This list is subject to change as the demands of the design create or eliminate needs of the system based on desired outcomes of the team.

- Create a magnification system between 3x and 5x
- Add an eyepiece to view the system through
- Create an optical system that intakes infrared light
- Create an infrared display
- Create a system which can turn the display off, on, or dimmed
- Create a photodetection system which communicates with our infrared display
- Have this system dim or brighten this display according to exterior illumination
- Mount an Infrared LED
- Mount/create a range finder
- Create a system to display that range
- Design a multi-rail DC-DC power distribution network that converts battery voltage to the specific voltages required by the single-board computer, the IR sensor, and the micro display
- Meet a hard power budget ceiling to ensure continuous runtime for an 1+ hours
- Route a custom, layered printed circuit board in an EDA tool that complies with tight geometric constraints to fit inside a handheld optic housing.
- Ensure proper high-speed trace routing and noise isolation between the digital video lines feeding the micro display and the sensitive analog lines of the ambient light sensor.
- Select and interface an ambient light sensor with an appropriate analog-to-digital converter to capture smooth gradations
- Program the microprocessor to ingest raw video frames from the IR sensor at a minimum frame rate of 30 frames per second
- Develop software/hardware for sensor acquisition, image switching/blending, and system control.
- Account for component placement, optical alignment, heat generation, power requirements, and user accessibility.
- Ensure transition stability, latency, image artifacts, and behavior near the switching/blending thresholds.

Stretch Objectives

- Implement a battery management system circuit featuring overvoltage/undervoltage protection
- Implement a software-defined low-pass filter that ensures the visual-to-IR transition responds to environmental shifts but ignores instantaneous noise
- Build a laser range finder
- Display distance to object indicated by range finder
- Create manually adjustable lens system

2.4 Features/Functionalities

The features listed are what a user may demand of the product. Based on these features, the design of the product can take shape. The main objective of the device is to see in the visible and IR spectrums; however, users will want other features that make the device friendly to manage. On top of the base features, users want simple interfaces to command the device, easy to use focusing features, easy to change batteries, and decent battery life. Below is a list of features we want to include in our device. This list is subject to change based on the needs of the device; more features could be added to help with user experience.

- Portable
- Magnifies
- Day and night view
- IR light
- Easy focusing
- Automatic illumination based on ambient light
- Pre-programmed smart controls
- Easy battery change

2.5 Table of 7 engineering specifications

The below list is a set of objective metrics we would like our device to meet. Meeting these specifications would allow the device to meet the required features our potential users

would want in a device of this nature. The first three rows are what should be demonstrated during the final demo. The device should be small enough dimensionally and weight to be handheld or mounted to a stand. The device should intake IR light to create an image even in dark conditions. In high light conditions, it should not react to IR light. The optic should have an FOV of 3 meters at a 30 meter distance.

Table 1: Overall System Specifications of Universal Knight Vision Scope

Parameter	Target Values	Reasoning
Magnification/Magnification range (through field of view)	The user will be able to magnify a field of view of a 3 meter diameter at 30 meters	Magnification = $\text{FOV}(\text{unmagnified}) / \text{FOV}(\text{magnified})$. A known object is first viewed by the eye without the magnifying optic, and the portion of the object visible within the viewer's field of view is measured. The object is then viewed through the optic, and the new field of view is measured under the same conditions. For a 3× magnification, the viewer should see approximately one-third of the original field of view, while a 5× magnification should reduce the visible field to approximately one-fifth of the original field of view.
Visible light sensitivity	Detecting visible wavelengths while ignoring IR wavelengths	The photodetection unit must react to visible light for the functionality of the device. If the device reacts to IR light, it could disrupt switching between day and night modes and turning on and off the IR illumination unit.
Near-Infrared range	800-1200 nm	Outside of human vision while still being close enough to use ambient light in the darkness. This is demonstrated by projecting an image onto the screen even in darkness.
Display magnified image on ocular lens in high light conditions	6 mm x 6 mm	The user must be able to see an image on the eyepiece in normal room lighting conditions where they can distinguish a human standing 20 feet away from the optic.

An IR image is projected on ocular lens in low-light conditions	6 mm x 6 mm	Once low light conditions are established, the viewer must observe an IR image on the eyepiece of identical magnification to the high light image within 10 seconds of the ambient light condition change.
Dimensions	<18"x10"x10"	Large enough to house electronics and optics while still being small enough to not be cumbersome
Weight	<5lb	Lightweight so that it can easily be held to the eye without a tripod
Display brightness	Variable	Optic dims and brightens based on ambient light conditions
Battery Life	+12 hours	Enables the user to continually use device through entire night
System power consumption	<3.5 watts	The total power drawn across all voltage rails during peak operation remains stable.
Thermal Management	<45° C	Enclosure surface temperature must remain below during continuous indoor operation to ensure safe handheld use.

Table 2: Component Specifications

Component	Parameter	Target Values	Description
Infrared Light	Range	750-950 nm	The light must have enough range to gather enough information on the surrounding area
Ambient light Sensor	Range	450-650 nm	The sensor must be good enough to ensure the device transitions exactly when the human

			eye naturally struggles to see.
Micro display	Resolution	30 frames per second	The display needs to be clear enough for the user to be able to see clearly
Camera	Image Delay	<100 ms	The camera must be able to assess and transfer information fast enough to the user.

2.6 Hardware block diagram

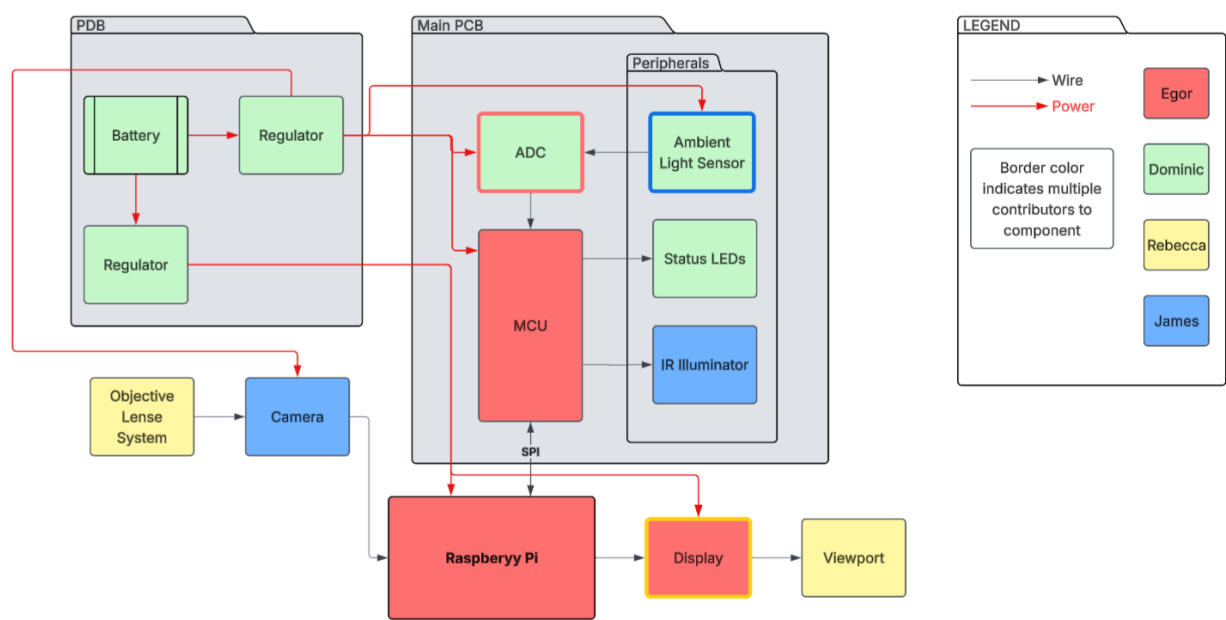
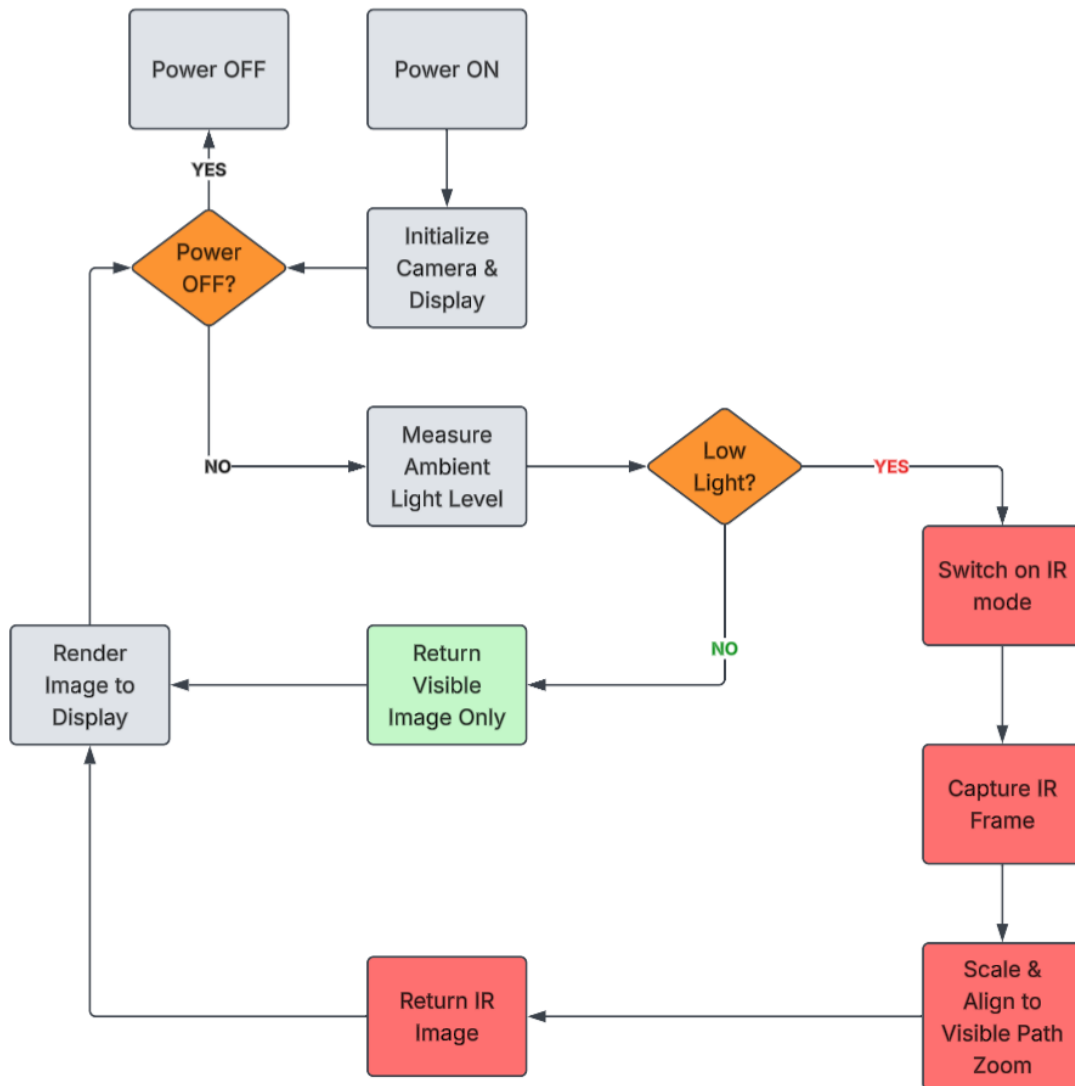


Figure 1: U.K.V.S hardware block diagram. This diagram displays the core components needed for the optical device to process captured frames as well as provide illumination and display device stats. Contributors are separated by color with colored borders if sharing components.

2.7 Software diagram/flowchart



The software flowchart outlines the process that the U.K.V.S device follows to process and relay images to the display including an ambient light checker to autonomously switch between visible and IR imaging.

Figure 2: Software Flowchart by Authors

2.8 Prototype illustration/blueprint

The figure below illustrates a side view of our initial plan for the optic. We plan to mount an IR LED and a phototransistor on top of the optic to provide illumination as needed and to detect ambient light, respectively. We plan to have an objective, magnifying, and relay lens behind the aperture leading to an IR-enabled camera, which is sensitive to both IR and visible light. We plan to place an IR filter in front of it that can be removed in low-light conditions. The IR camera intake should be relayed to a display, which will be projected onto an ocular lens. We will include a command center with electronic hardware to speak to the phototransistor, IR LED, IR camera, and display.

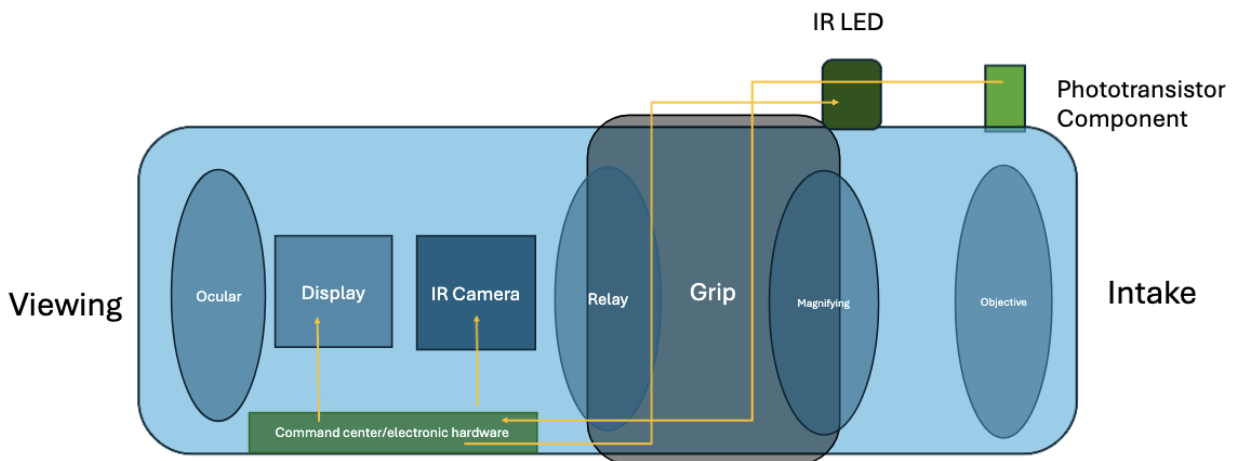


Figure 3: Side view of optic, illustrating the initial lens system design, IR components, and electrical hardware that will be used to control the software components of the system.

The figure below illustrates our initial plan for the optic, looking down the front of the intake side of the optic. It shows the alignment of the phototransistor, IR LED, and objective lens from the front. The phototransistor will sit in front of the LED. The objective lens is the first visible lens when looking down the front of the optic. The other three lenses, the camera, the display, and the command center sit behind it in the order shown in Figure 3.

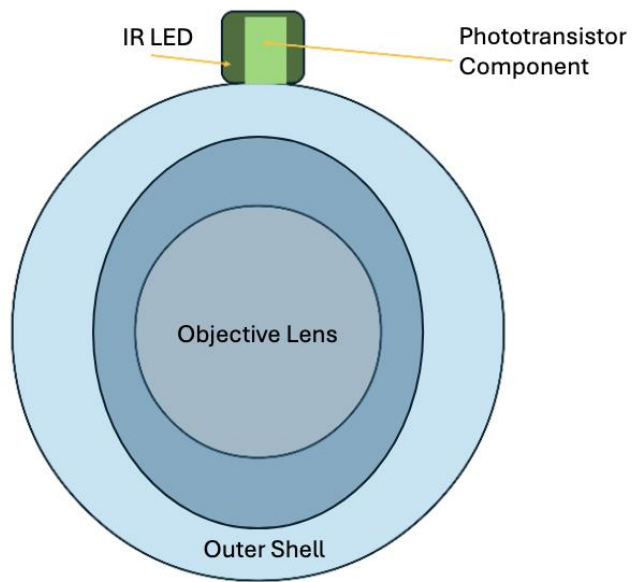


Figure 4: Front view of the optic, offering a more complete representation of the optic's overall geometry. This view also illustrates the alignment between the IR LED and phototransistor, as well as the position of the first lens within the optical system.

2.9 House of quality

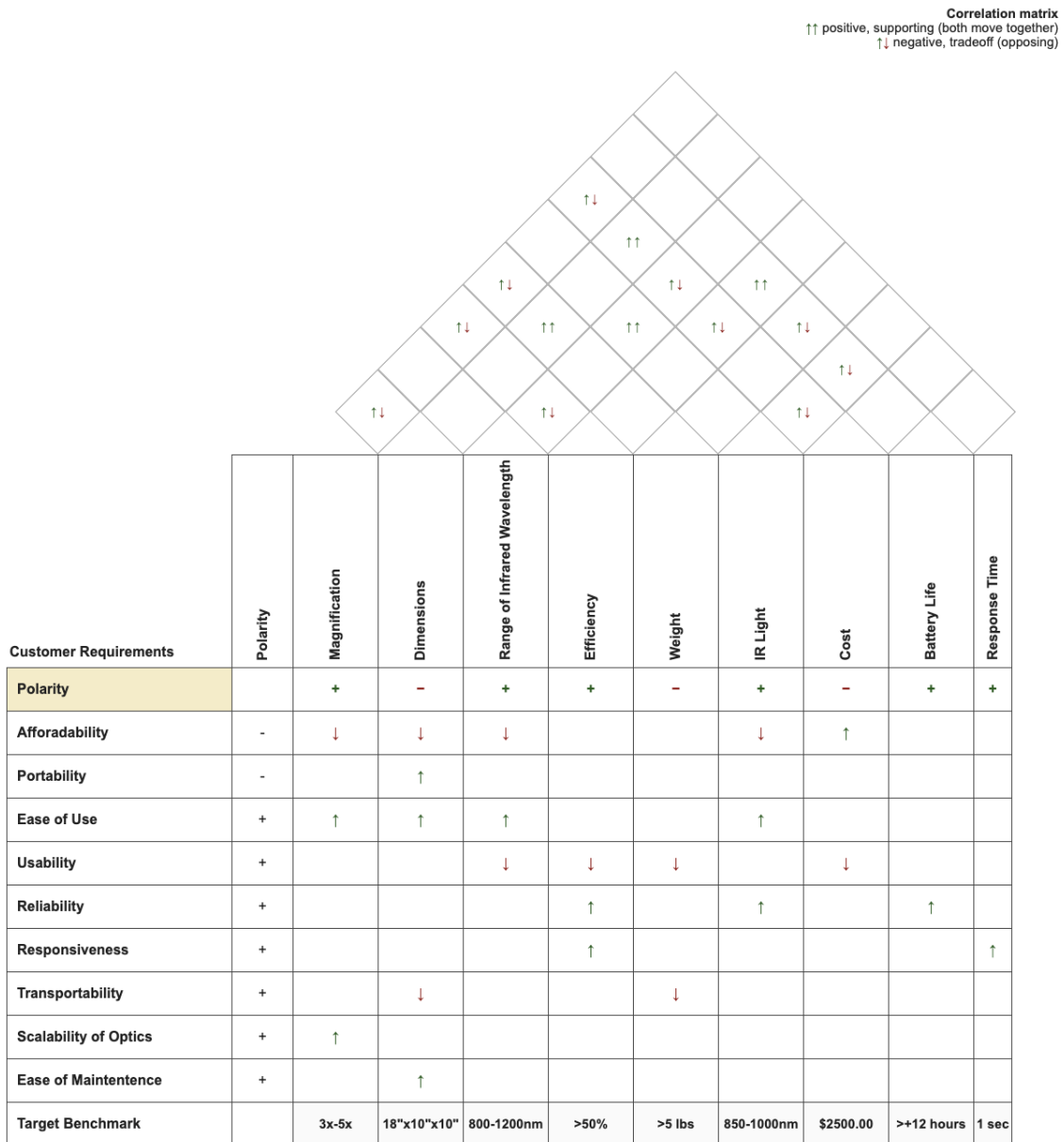


Figure 5: House of Quality, indicating relationships between engineering requirements and customer desires

Chapter 11: Administrative Contents

11.1 Budget and financing

Our goal for the operation budget is under \$2,500.00. Some components are less expensive but will require more of them. Barring a very generous benefactor, we will be footing these costs ourselves with those costs equally divided among the 4 group members over the course of two semesters. Given the financial restraints of students, these components will not be purchased at once, but over the course of 2 semesters as needed. Hopefully, no student spends more than \$250 per semester over the course of 2 semesters. Generic items may be purchased and provided by UCF.

Table 3: Necessary items and cost

Component Number	Description	Cost Estimate	Quantity	Procurement
N-BK7 and N-SF11 Bi-Convex Lenses	Lenses for the optical magnification system	\$45	~ 4	ThorLabs
Raspberry Pi Global Shutter Camera	Camera to collect IR and visible light	\$65	1	SparkFun Electronics
Photo transistor	Detect ambient light	\$12	Pack of 20	Amazon
Raspberry Pi 5	A single board computer with four cores that includes USB ports.	\$180 - \$305	1	SparkFun Electronics
MCP3008	Analog to Digital Converter	\$2.90	1	DigiKey
STM32G431CBT6	MCU	\$7.04	1	DigiKey
IR reject filter	650-940 IR band reject filter	\$10	1	Amazon
PCB	A printed circuit board	\$70.00	1	Jlcpcb.com

	used to mechanically support and electrically connect electronic components			
Misc.	Other components which are currently unforeseen but will almost certainly become necessary/ and or small components such as wires, resistors, etc	\$200	As needed	As needed
Total		>\$1,000.00		

11.2 Project milestones/plan

Below is a framework for when deadlines are and which items should be handled by each group member of the design team. Semester 1 has more structure ,while Semester 2 is more open ended to accommodate corrections and changes as necessary. The table is a schedule of events while the Gantt chart describes each member's tasks and when they should be delivered.

Table 4: Project plan deadlines

Milestone	Due Date
Submit Divide & Conquer	09/11/26
Divide & Conquer Revision	09/18/26
Acquire parts for midterm demo and begin subsystem construction. Obtain phototransistor and circuit elements. Begin fabrication.	10/01/26
Complete construction and preliminary testing of the SD1 Midterm demo.	10/26/26

CREOL SD1 Midterm Demo - verify phototransistor response and software communication.	10/28/26
Midterm Milestone Report	11/06/26
Acquire parts for final demo and begin subsystem construction. Obtain IR filter/switching components. Begin fabrication and testing of the SD1 final demo subsystem.	11/11/26
Midterm Report Revision	11/13/26
Finish building and testing final demo. Verify that the IR filter functions and can be flipped on and off via software.	11/30/26
CREOL SD1 Final Demo	12/02/26
Final SD1 Report + Mini demo video	12/03/26
Acquire all materials for SD2 project assembly, primary lenses to build the optical system, an IR camera, and a display. Plan to 3D print a shell to contain our entire system	2 nd week of Spring Semester
Begin SD2 project assembly. Focus on SD2 demo but work on all components	2 nd week of Spring Semester
Finish building for SD2 midterm demo. Verify that lens system will project an image onto an area the size of our camera sensor. Print shell if this has not already been done.	Week 8 Spring 2027
SD2 Midterm Demo	Week 9 Spring 2027
Testing and troubleshooting SD2 Project. The project should be in its final stages at this point: take care of final issues and assemble into shell.	Week 9 – 13 Spring 2027
Finish Building SD2 Project. Continue to test and refine issues.	3 days before deadline
SD2 Project due	End of Spring 2027

Universal Knight Vision Scope

Group 5

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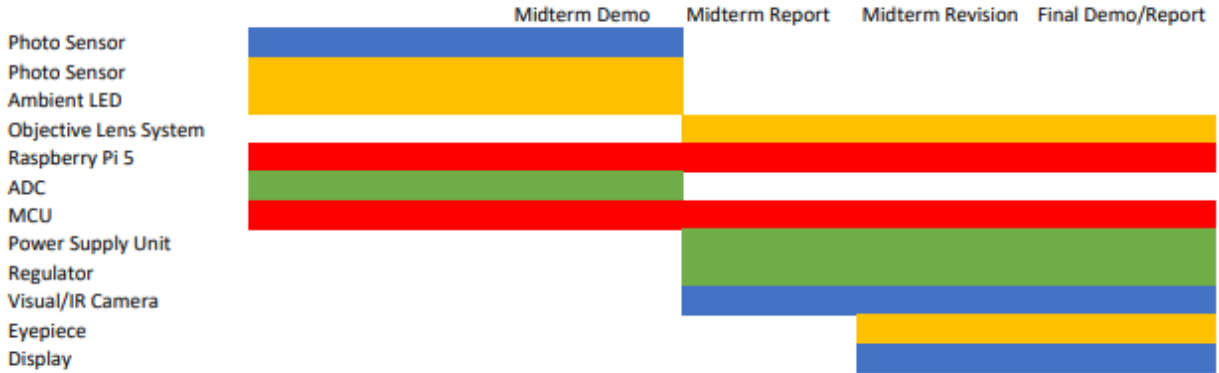
Dominic Alexis

Egor Shevchanko

All

Project Milestones and Schedule

SD1



SD2

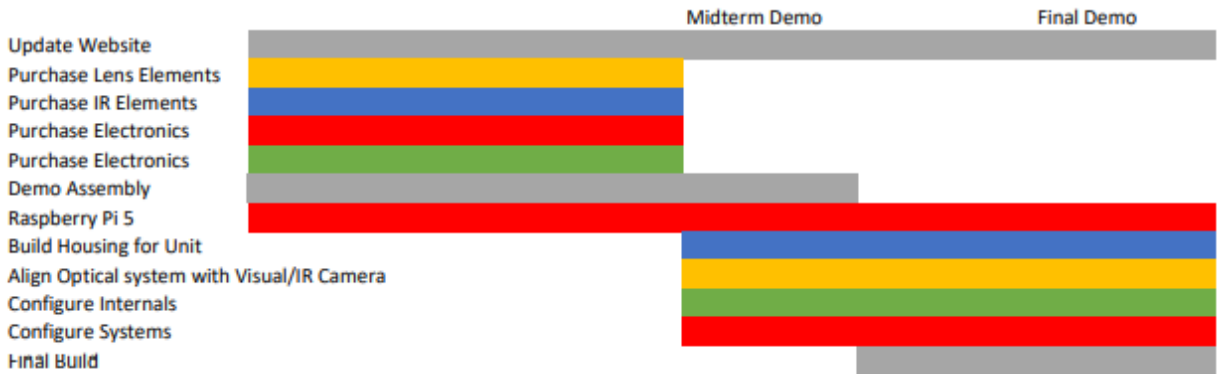


Figure 6: Milestone and task flowchart

11.3 Risk Assessment

The below table identifies risks to the project, how likely they are to happen, impact to the project, how we plan to avoid the given risk, and when we plan to demonstrate to our client or internally as a group that we have resolved the issue.

Table 4: Risk assessment and mitigation

Identified Risk	Likelihood	Impact	Mitigation Plan	Risk Retirement
The photodetection system cannot	High	High – we want an automatic system to	Create a circuit with the phototransistor	CREOL SD1 Midterm

detect ambient illumination levels accurately or cannot communicate this information to a software which will trigger changes to display brightness		trigger the infrared display. However, if it fails, we could have a manual switching system instead	we plan to use on our project and link it to an IR LED. Once ambient room light is turned off, the IR light LED should turn in within 10 seconds.	Demo (week 10)
Failure of a system to filter or not filter IR light	Low	High – the optic must intake IR light in low light conditions, or else it will not function at night. Filtering IR light in high light conditions will protect the sensors, improving device longevity.	Demo an IR filter with a mechanical switch controlled by software. Demonstrate that the filter does not allow IR light to pass and that the mechanical switching happens within 10 seconds of the switch command being sent	CREOL SD1 Final Demo (week 15)
Failure of lens system to project an image into a 6mm x 6mm camera lens clearly	Moderate	Catastrophic – if unable to project an image onto the camera, this optic will not be able to display images	Build a test lens setup to project an image onto 6mm x 6mm surface. The image shall cover at least 5mm x 5mm of the surface and not extend more than 1mm x 1mm past the edges of the surface.	CREOL SD2 Midterm Demo (week 9)
The photodetection	High	Catastrophic – if the IR light	Have a phototransistor	SD2 final demo – related

system may not trigger our display on once the IR light source turns on because the device will believe the IR light is external illumination		source triggers the IR display off, our night vision optic will not function	attached to a screen. It should be off when illuminated with visible light, but in when illuminated with IR light	to Demo 2 as the filter eliminates this risk
The optic contains high-speed sensors and switching DC-DC regulators, significant electromagnetic interference, which means high-frequency noise will be present on the PCB which might make the sensor readings fluctuate wildly.	High	Moderate – Causes the display to flicker	Keep the light sensor traces as short as possible. Connect the sensors to the PCB and observe the sounds heard.	SD2 final demo
Tight physical constraints mean the custom PCB will be enclosed in a small, unventilated space. Power dissipation from voltage regulators, charging circuits, and the primary processor can quickly heat up the sealed enclosure.	Moderate	High - Reduces battery life drastically, introduces thermal noise into the IR sensor, and the device could short and combust.	Design thermal vias or small passive heatsink on the PCB underneath hot components (to draw heat toward an external metal chassis.	SD2 final demo

Designing a multi-layered PCB with fine components introduces a high chance of manufacturing defects and components not being available.	Moderate	Catastrophic – This would leave us with faulty or no hardware to test	Build a modular prototype using breakout breadboards and jumper wires.	SD2 final demo
Latency impeding usability – past a certain threshold (50ms+) the device will start lagging and cause disorientation	Moderate	High – this will negatively impact user experience and will	Utilize multiprocessing to dedicate one CPU core to capturing frames from visible sensor, one core for capturing frames from near-IR sensor, and another core to process frames	SD2 final demo
Incorrect scaling/alignment of IR image over visible image magnification	Moderate	Catastrophic – device would fail the usability requirement if providing an overlay that does not reflect reality	Get the visible path's actual magnification and field of view from Zemax results and derive correction math from those values	SD2 final demo
Single point of failure in the processor	Moderate	High - A crash during IR use could take down the visible path too, losing all vision through the device	Use exception handling to catch errors seen by code and fall back to visible imaging. Additionally implement a watchdog timer that resets the processor if the	SD2 final demo

			main loop stops responding within some window, used when exception handling fails	
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11.4 Conclusion

We intend to build a dual night- and day- vision optic, which will work as a simple magnifier in high light conditions, while automatically transitioning to an infrared view in lower-light conditions. This optic should provide maximum convenience by offering a viewing aid in any light conditions, while being intuitive to use.

Appendices

Appendix A

Citations

OpenAI. (2026). *ChatGPT* (GPT-5.6 Sol) [Large language model]. Used Chat GPT to research the history of infrared optics

OpenAI. (2026). *ChatGPT* (GPT-5.6 Sol) [Large language model]. Used Chat GPT to edit history and goal sections of this proposal

Anthropic. (2026). *Claude* (Sonnet 5) [Large Language Model]. Used Claude to research electrical components

Appendix B

No copyright material used